Filename: BIOL328-HW3

Biology 328 - Homework Assignment 3
Student Name: ________________ | Date: Fall 2024

Developmental Biology Problem Set

Problem 1

Describe the process of gastrulation in vertebrate embryos. Include:

- The formation of the three germ layers
- Key signaling pathways involved
- Regional differences along the anterior-posterior axis

Answer:

Gastrulation is a critical phase in early embryonic development where the blastula reorganizes into a three-layered structure called the gastrula. During this process, cells migrate to form the three primary germ layers: ectoderm (outer layer), mesoderm (middle layer), and endoderm (inner layer).

The process begins with the formation of the primitive streak in birds and mammals, or the blastopore in amphibians and fish. Cell movements including invagination, involution, and epiboly coordinate to position cells in their appropriate germ layers.

Key signaling pathways involved include:

- BMP (Bone Morphogenetic Protein) signaling for mesoderm induction
- Nodal signaling for establishing left-right asymmetry
- Wnt signaling for anterior-posterior patterning
- Notch signaling for cell fate decisions

Regional differences along the anterior-posterior axis are established through concentration gradients of morphogens. The anterior region gives rise to head structures, while the posterior region forms tail structures. Hox genes play a crucial role in specifying segment identity along this axis.

Problem 2

Explain the concept of morphogenetic fields and how they contribute to pattern formation during development. Provide an example.

Answer:

Morphogenetic fields are regions of the embryo that have the potential to develop into specific structures or patterns. These fields are defined by gradients of signaling molecules (morphogens) that influence cell fate based on their position and concentration.

A classic example is the limb bud development in vertebrates. The limb bud contains two major organizing centers:

- **Apical Ectodermal Ridge (AER):** Maintains the progress zone and controls proximal-distal patterning (shoulder to finger tip)
- **Zone of Polarizing Activity (ZPA):** Establishes anterior-posterior patterning (thumb to pinky) through Sonic hedgehog (Shh) signaling

Cells interpret their position within these morphogenetic fields based on morphogen concentration thresholds, leading to activation of specific transcription factors and ultimately determining the final structure that develops in that region.

Problem 3

Compare and contrast symmetric and asymmetric cell division in developmental biology. When is each type important?

Answer:

Symmetric cell division produces two identical daughter cells with the same developmental potential. This is important during early cleavage stages when rapid cell proliferation occurs without differentiation, and later for expanding specific cell populations.

Asymmetric cell division produces two daughter cells with different fates. This is crucial for generating cellular diversity and occurs when:

- Stem cells divide to produce one stem cell (self-renewal) and one differentiating cell
- Neural progenitor cells generate different neuronal types
- Establishing polarity in tissues

Asymmetric division involves the unequal distribution of cell fate determinants (proteins, RNAs) or signaling molecules, ensuring that one daughter cell receives factors that will direct it toward a different developmental path than its sibling.

Problem 4

Describe the role of homeotic genes in body plan development. What happens when these genes are mutated?

Answer:

Homeotic genes are master regulatory genes that control the identity of body segments and structures. They encode transcription factors that activate or repress downstream target genes, determining what structure will develop at a particular location along the body axis.

The most well-studied homeotic genes are the Hox genes, which are arranged in clusters on chromosomes in the same order as the body segments they control (colinearity rule).

When homeotic genes are mutated:

- **Loss-of-function mutations:** Can cause structures to develop with the identity of adjacent segments (homeosis). For example, a Drosophila fly with a mutated Antennapedia gene develops legs where antennae should be.
- **Gain-of-function mutations:** Can cause structures to develop with the identity of different segments, such as developing wings where halteres should be.

These genes demonstrate the profound importance of temporal and spatial control of gene expression in establishing the correct body plan.